

WALMART-CASE-STUDY

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Note:-The purchases are converted in lakhs so do consider it and market is considered is India

1. What are the basic characteristics, structure, and statistical summary of the Walmart dataset?

ANS(CODE):

```
import numpy as np

import pandas as pd

data = pd.read_csv('walmart_data.csv')

print(data.info())

print(data.describe())

print(data.columns)

print(data.shape)

print(data.ndim)

print(data.isnull().sum())
```

OUTPUT:

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 550068 entries, 0 to 550067
Data columns (total 10 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   User_ID                               550068 non-null  int64
1   Product_ID                            550068 non-null  object
2   Gender                                550068 non-null  object
3   Age                                    550068 non-null  object
4   Occupation                            550068 non-null  int64
5   City_Category                          550068 non-null  object
6   Stay_In_Current_City_Years            550068 non-null  object
7   Marital_Status                         550068 non-null  int64
8   Product_Category                       550068 non-null  int64
9   Purchase                              550068 non-null  int64
dtypes: int64(5), object(5)
```

	User_ID	Occupation	Marital_Status	Product_Category
count	5.500680e+05	550068.000000	550068.000000	550068.000000
mean	1.003029e+06	8.076707	0.409653	5.404270
std	1.727592e+03	6.522660	0.491770	3.936211
min	1.000001e+06	0.000000	0.000000	1.000000
25%	1.001516e+06	2.000000	0.000000	1.000000
50%	1.003077e+06	7.000000	0.000000	5.000000
75%	1.004478e+06	14.000000	1.000000	8.000000
max	1.006040e+06	20.000000	1.000000	20.000000

```

      Purchase
count  550068.000000
mean    9263.968713
std    5023.065394
min      12.000000
25%    5823.000000
50%    8047.000000
75%   12054.000000
max    23961.000000
Index(['User_ID', 'Product_ID', 'Gender', 'Age', 'Occupation', 'City_Category',
      'Stay_In_Current_City_Years', 'Marital_Status', 'Product_Category',
      'Purchase', 'Occupation_category', 'Marital_category'],
      dtype='object')
(550068, 12)
2
```

```

User_ID      0
Product_ID   0
Gender       1
Age          1
Occupation   1
City_Category 1
Stay_In_Current_City_Years 1
Marital_Status 1
Product_Category 1
Purchase     1
dtype: int64

```

2. Add the necessary columns and be added to the Walmart dataset for analysis also remove the null values from the original dataset?

ANS(CODE):

```

import pandas as pd
import numpy as np

data = pd.read_csv('walmart_data.csv')

occupation_map = {
    0: 'Occupation 0',
    1: 'Occupation 1',
    2: 'Occupation 2',
    3: 'Occupation 3',
    4: 'Occupation 4',
    5: 'Occupation 5',
    6: 'Occupation 6',
    7: 'Occupation 7',
    8: 'Occupation 8',
    9: 'Occupation 9',
    10: 'Occupation 10',
    11: 'Occupation 11',
    12: 'Occupation 12',
    13: 'Occupation 13',
    14: 'Occupation 14',
    15: 'Occupation 15',

```

```

16: 'Occupation 16',
17: 'Occupation 17',
18: 'Occupation 18',
19: 'Occupation 19',
20: 'Occupation 20'
}

married_map = {
    0: 'Single',
    1: 'Married'
}

data['Occupation_category'] = data['Occupation'].map(occupation_map)
data['Marital_category'] = data['Marital_Status'].map(married_map)

print(data.head(5))

data = data.dropna()

print(data.isnull().sum())

data.to_csv('walmart_data.csv', index=False)

```

OUTPUT:

```

User_ID Product_ID Gender Age Occupation City_Category \
0 1000001 P00069042 F 0-17 10 A
1 1000001 P00248942 F 0-17 10 A
2 1000001 P00087842 F 0-17 10 A
3 1000001 P00085442 F 0-17 10 A
4 1000002 P00285442 M 55+ 16 C

Stay_In_Current_City_Years Marital_Status Product_Category Purchase \
0 2 0 3 8370
1 2 0 1 15200
2 2 0 12 1422
3 2 0 12 1057
4 4+ 0 8 7969

Occupation_category Marital_category
0 Occupation 10 Single
1 Occupation 10 Single
2 Occupation 10 Single
3 Occupation 10 Single
4 Occupation 16 Single

```

```

User_ID 0
Product_ID 0
Gender 0
Age 0
Occupation 0
City_Category 0
Stay_In_Current_City_Years 0
Marital_Status 0
Product_Category 0
Purchase 0
Occupation_category 0
Marital_category 0
dtype: int64

```

- How do customer purchasing patterns vary across different city categories, product categories, and occupations?

ANS(CODE):

```
import numpy as np

import pandas as pd

data = pd.read_csv('walmart_data.csv')

# Number of unique users per city category

number_of_users_per_city_type = data.groupby('City_Category')['User_ID'].nunique()

# Number of different products per product category

number_of_products_per_product_category =
data.groupby('Product_Category')['Product_ID'].nunique()

# Amount spent across different product categories

amount_per_product_category_spent = data.groupby('Product_Category')['Purchase'].sum()

# Output

print("The number of users per city:\n", number_of_users_per_city_type)

print("The number of products per category:\n",number_of_products_per_product_category)

print("The amount spent per product category:\n",round(amount_per_product_category_spent
/ 100000, 3))
```

OUTPUT:

```

The number of users per city:
City_Category
A    1045
B    1707
C    3139
Name: User_ID, dtype: int64
The number of products per category:
Product_Category
1         493
2         152
3          90
4          88
5         967
6         119
7         102
8        1047
9           2
10          25
11         254
12          25
13          35
14          44
15          44
16          98
17          11
18          30
19           2
20           3
Name: Product_ID, dtype: int64

```

```

The amount spent per product category:
Product_Category
1        19100.138
2         2685.162
3         2040.847
4          273.805
5         9418.352
6         3241.503
7          608.967
8         8543.188
9           63.703
10        1008.373
11        1137.911
12           53.318
13           40.086
14          200.147
15          929.690
16        1451.206
17           58.787
18          92.902
19            0.594
20            9.447
Name: Purchase, dtype: float64

```

4. How does customer spending vary across different occupations, marital categories, and years of residence in the current city?

ANS(CODE):

```

import numpy as np

import pandas as pd

data = pd.read_csv('walmart_data.csv')

amount_spent_per_occupation = data.groupby('Occupation')['Purchase'].sum()

# Amount spent per occupation

amount_spent_across_marital_category = data.groupby('Marital_category')['Purchase'].sum()

# Amount spent across different marital categories

amount_spent_as_per_living_in_city =
data.groupby('Stay_In_Current_City_Years')['Purchase'].sum()

```

```
# Amount spent based on years of residence in the current city

# Output

print("The amount spent per occupation:\n",

      round(amount_spent_per_occupation / 100000, 3))

print("The amount spent across marital categories:\n",

      round(amount_spent_across_marital_category / 100000, 3))

print("The amount spent based on years of residence in the current city:\n",

      round(amount_spent_as_per_living_in_city / 100000, 3))
```

OUTPUT:

```
The amount spent per occupation:
Occupation
0      6354.070
1      4246.141
2      2380.286
3      1620.022
4      6662.445
5      1136.498
6      1884.168
7      5573.716
8       147.374
9       543.400
10     1158.445
11     1067.516
12     3054.494
13       719.195
14     2594.547
15     1189.602
16     2383.470
17     3932.815
18       607.215
19     737.006
20     2965.704
Name: Purchase, dtype: float64
The amount spent across marital categories:
Marital_category
Married      20868.853
Single       30089.274
Name: Purchase, dtype: float64
```

```
The amount spent based on years of residence in the current city:
Stay_In_Current_City_Years
0      6829.792
1     17928.725
2      9491.739
3      8849.027
4+      7858.844
Name: Purchase, dtype: float64
```

5. How do customer demographics and spending patterns vary across gender, age groups, and marital status?

ANS(CODE):

```

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

data = pd.read_csv('walmart_data.csv')

# Basic gender count
gender_count = data.groupby('Gender')['User_ID'].count()

print(gender_count)

# Separate male and female data
female = data[data['Gender'] == 'F']
male = data[data['Gender'] == 'M']

# Number of users by age for each gender
female_count = female.groupby('Age')['User_ID'].count()
male_count = male.groupby('Age')['User_ID'].count()

# Number of married/unmarried users by age for each gender
female_marriage_status = female.groupby(
    ['Marital_Status', 'Age']
)['User_ID'].count()

male_marriage_status = male.groupby(
    ['Marital_Status', 'Age']
)['User_ID'].count()

print("The female marriage status:\n", female_marriage_status)
print("The male marriage status:\n", male_marriage_status)

# Spending by marital status for females
spending_per_marital_status_per_female = \
    female.groupby('Marital_Status')['Purchase'].sum()

spending_per_marital_status_per_female.index = \
    spending_per_marital_status_per_female.index.map({
        0: 'Unmarried',
        1: 'Married'
    })

```

```

    })

# Spending by marital status for males

spending_per_marital_status_per_male = male.groupby('Marital_Status')['Purchase'].sum()

spending_per_marital_status_per_male.index =
spending_per_marital_status_per_male.index.map({
    0: 'Unmarried',
    1: 'Married'
})

print(
    "Spending as per marital status for female:\n",
    round(spending_per_marital_status_per_female / 100000, 3)
)

print(
    "Spending as per marital status for male:\n",
    round(spending_per_marital_status_per_male / 100000, 3)
)

# Male vs female users by age

plt.plot(
    male_count.index,
    male_count.values,
    color='black',
    label='Male'
)

plt.plot(
    female_count.index,
    female_count.values,
    color='red',
    label='Female'
)

```

```

plt.xlabel('Age-group')
plt.ylabel('User count')
plt.title('(Age-group)-(User-count)')
plt.legend()
plt.show()

# Married/unmarried users by age for females and males

plt.plot(
    female_marriage_status.xs(0, level='Marital_Status'),
    color='red',
    label='Female - Unmarried'
)

plt.plot(
    female_marriage_status.xs(1, level='Marital_Status'),
    color='purple',
    label='Female - Married'
)

plt.plot(
    male_marriage_status.xs(0, level='Marital_Status'),
    color='black',
    label='Male - Unmarried'
)

plt.plot(
    male_marriage_status.xs(1, level='Marital_Status'),
    color='brown',
    label='Male - Married'
)

plt.xlabel('Age')
plt.ylabel('User Count')
plt.title('User Count by Age, Gender and Marital Status')
plt.legend()

```

```
plt.show()
```

OUTPUT:

Gender

F 135809

M 414259

Name: User_ID, dtype: int64

The female marriage status:

	Marital_Status	Age
0	0-17	5083
	18-25	18357
	26-35	30078
	36-45	16649
	46-50	3166
	51-55	3580
	55+	1908
1	18-25	6271
	26-35	20674
	36-45	10521
	46-50	10033
	51-55	6314
	55+	3175

Name: User_ID, dtype: int64

The male marriage status:

	Marital_Status	Age
0	0-17	10019
	18-25	60187
	26-35	103218
	36-45	49728
	46-50	9524
	51-55	7259
	55+	5975

1	18-25	14845
	26-35	65617
	36-45	33115
	46-50	22978
	51-55	21348
	55+	10446

Name: User_ID, dtype: int64

Spending as per marital status for female:

Marital_Status

Unmarried 6841.541

Married 5020.785

Name: Purchase, dtype: float64

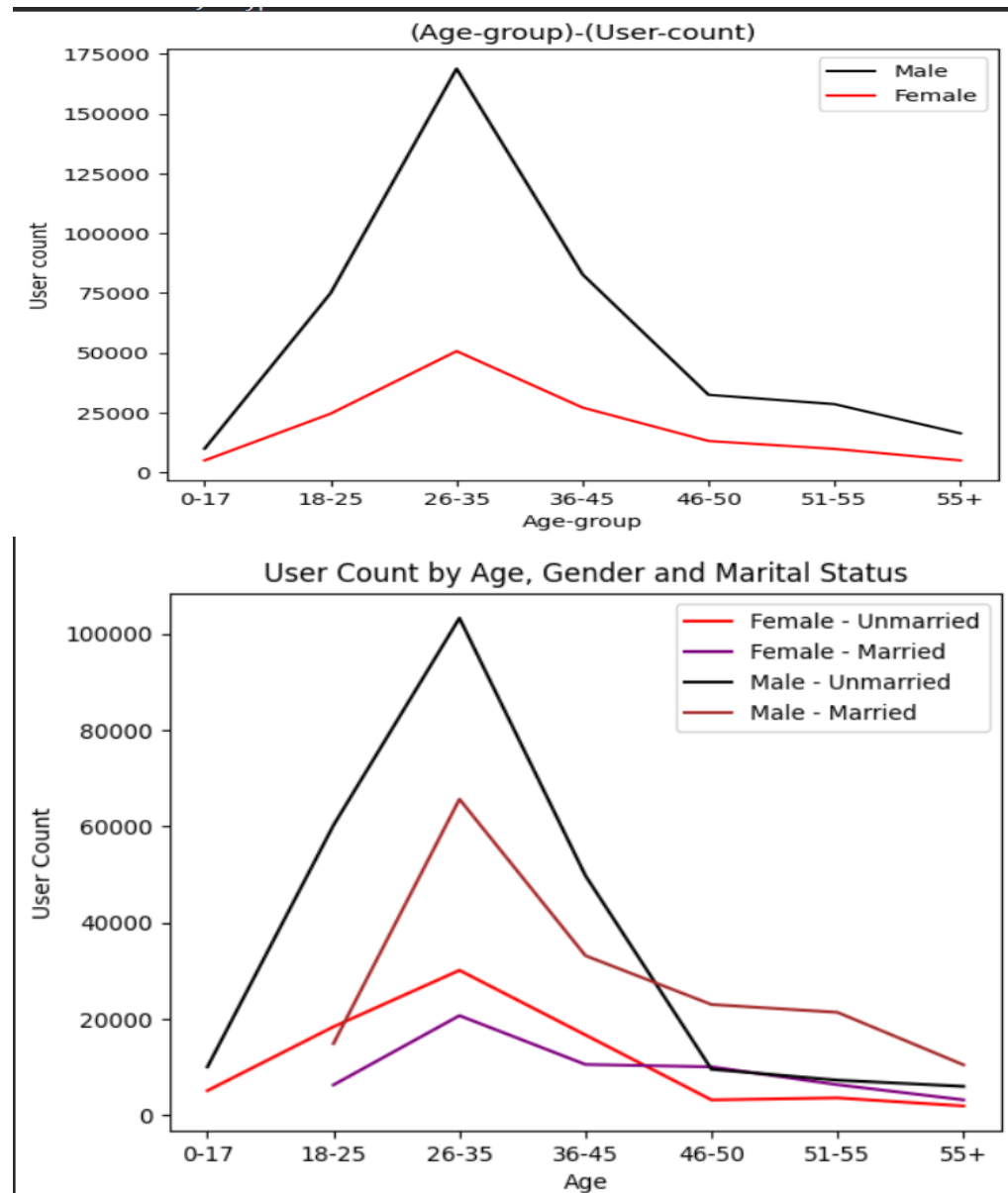
Spending as per marital status for male:

Marital_Status

Unmarried 23247.733

Married 15848.068

Name: Purchase, dtype: float64



6. How do customer spending patterns vary across city categories, age groups, years of residence, and product categories, and how does average spending per customer differ between cities?

ANS(CODE):

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('walmart_data.csv')

#city category
spending_per_city = data.groupby('City_Category')['Purchase'].sum()
```

```

per_product_category_per_city =
data.groupby(['Product_Category','City_Category'])['Purchase'].sum()

highest_product_per_city =
per_product_category_per_city.groupby('City_Category').idxmax().apply(lambda x: x[0])

lowest_product_per_city =
per_product_category_per_city.groupby('City_Category').idxmin().apply(lambda x: x[0])

print("the spending per city:\n",round(spending_per_city/100000,3))

print("The highest ordered product category per city:\n",highest_product_per_city)
print("The lowest ordered product category per city:\n",lowest_product_per_city)

# spending analysis based on years stayed in current city

spending_by_city_stay_years = data.groupby('Stay_In_Current_City_Years')['Purchase'].sum()

spending_by_city_stay_years = round(spending_by_city_stay_years/100000,3)

print("Spending by years stayed in current city:\n", spending_by_city_stay_years)

#spending analysis based on the age

spending_by_age = data.groupby('Age')['Purchase'].sum()

spending_by_age = round(spending_by_age / 100000, 3)

print("Spending by age group:\n", spending_by_age)

plt.bar(spending_by_age.index, spending_by_age.values)

plt.xlabel('Age Group')

plt.ylabel('Spending (in lakhs)')

plt.title('Spending by Age Group')

plt.show()

#the average spending of customers per city

customer_count_by_city = data.groupby('City_Category')['User_ID'].nunique()

spending_per_customer = (

data.groupby('City_Category')['Purchase'].sum()

/ customer_count_by_city)

print("Average spending per customer by city:\n",round(spending_per_customer, 3))

#product categoryspending

spending = data.groupby('Product_Category')['Purchase'].sum()

spending = round(spending/100000,3)

plt.bar(spending.index, spending.values)

```

```
plt.xticks(spending.index)
plt.xlabel('Product Category')
plt.ylabel('Spending (in lakhs)')
plt.title('Spending by Product Category')
plt.show()
```

OUTPUT:

the spending per city:

City_Category

A 13164.717

B 21155.336

C 16638.075

Name: Purchase, dtype: float64

The highest ordered product category per city:

City_Category

A 1

B 1

C 1

Name: Purchase, dtype: int64

The lowest ordered product category per city:

City_Category

A 19

B 19

C 19

Name: Purchase, dtype: int64

Spending by years stayed in current city:

Stay_In_Current_City_Years

0 6829.792

1 17928.725

2 9491.739

3 8849.027

4+ 7858.844

Name: Purchase, dtype: float64

Spending by age group:

Age

0-17 1349.132

18-25 9138.487

26-35 20317.706

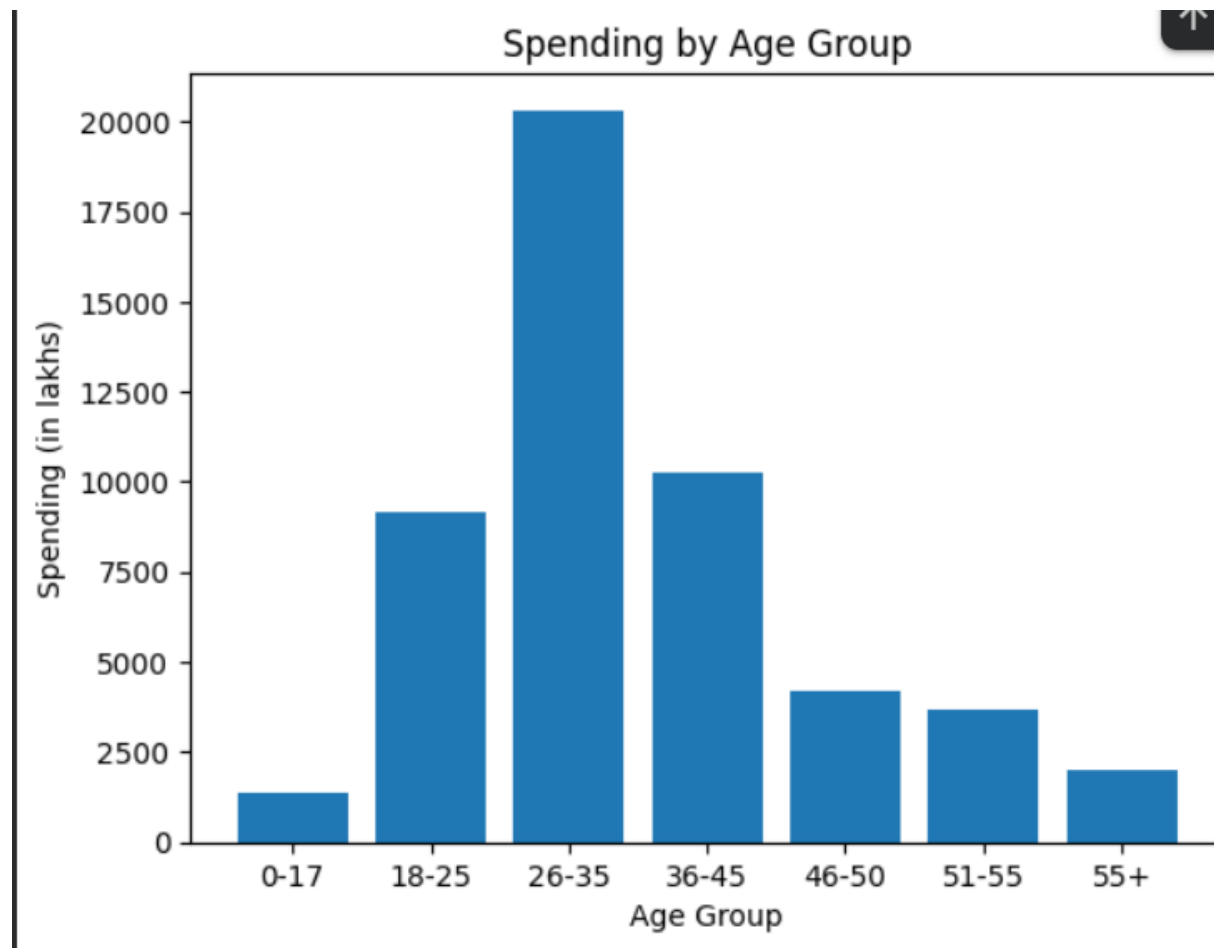
36-45 10265.699

46-50 4208.434

51-55 3670.996

55+ 2007.674

Name: Purchase, dtype: float64

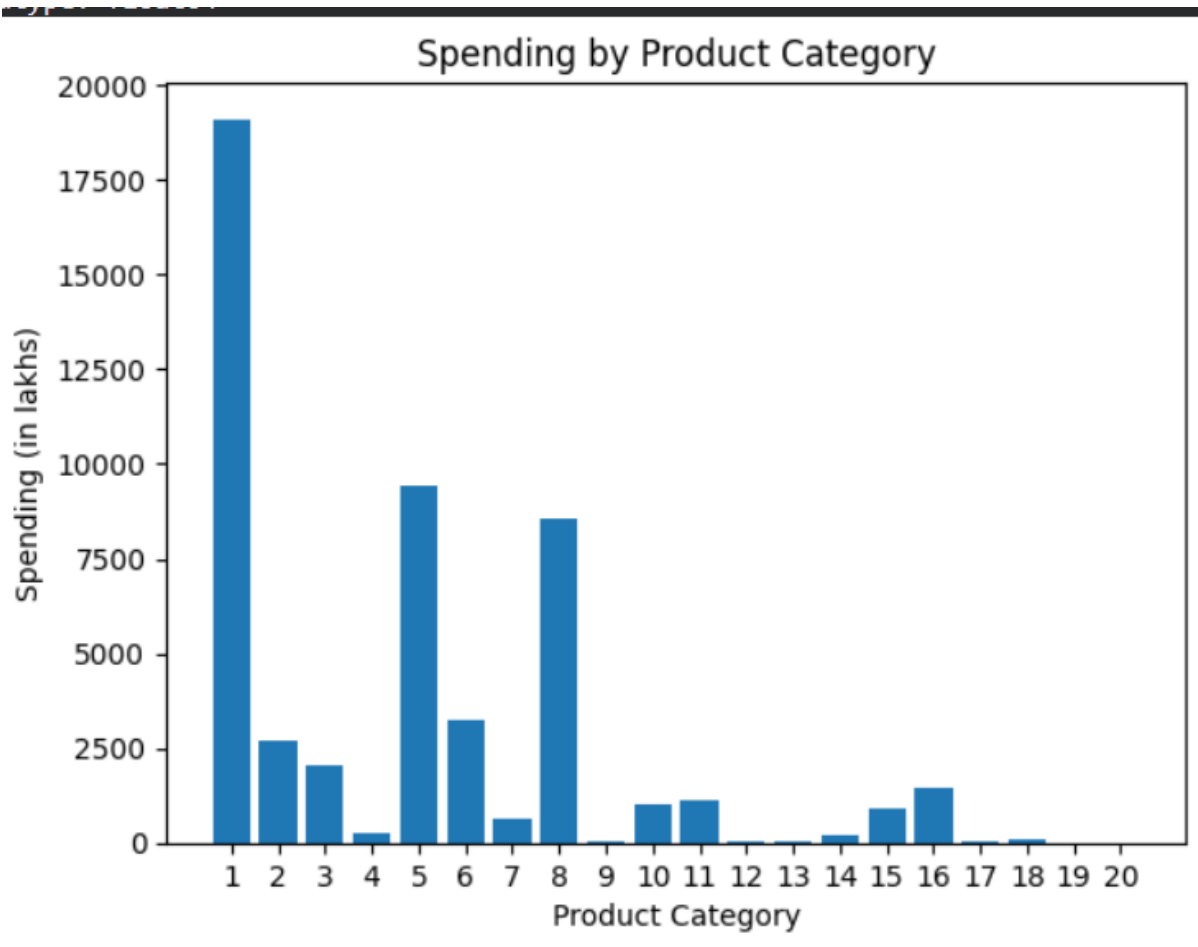


Average spending per customer by city:

A 1259781.494

B 1239328.415

C 530043.796



7. How does customer spending vary across different occupations, and how do spending patterns differ between male and female customers?

ANS(CODE):

```
import pandas as pd
import numpy as np

data = pd.read_csv('walmart_data.csv')

spending_per_occupation = data.groupby('Occupation_category')['Purchase'].sum()
spending_per_occupation = round(spending_per_occupation / 100000, 3)
print("spending_across_all_genders_occupation:\n", spending_per_occupation)

woman_based = data[data['Gender'] == 'F']
men_based = data[data['Gender'] == 'M']

woman_spending_per_occupation =
woman_based.groupby('Occupation_category')['Purchase'].sum()

men_spending_per_occupation = men_based.groupby('Occupation_category')['Purchase'].sum()
```

```
print("spending_of_woman_across_occupation:\n",  
      round(woman_spending_per_occupation / 100000, 3))  
print("spending_of_men_across_occupation:\n",  
      round(men_spending_per_occupation / 100000, 3))
```

OUTPUT:

spending_across_all_genders_occupation:

Occupation_category

Occupation 0 6354.070

Occupation 1 4246.141

Occupation 10 1158.445

Occupation 11 1067.516

Occupation 12 3054.494

Occupation 13 719.195

Occupation 14 2594.547

Occupation 15 1189.602

Occupation 16 2383.470

Occupation 17 3932.815

Occupation 18 607.215

Occupation 19 737.006

Occupation 2 2380.286

Occupation 20 2965.704

Occupation 3 1620.022

Occupation 4 6662.445

Occupation 5 1136.498

Occupation 6 1884.168

Occupation 7 5573.716

Occupation 8 147.374

Occupation 9 543.400

Name: Purchase, dtype: float64

spending_of_woman_across_occupation:

Occupation_category	
Occupation 0	1598.838
Occupation 1	1528.067
Occupation 10	328.036
Occupation 11	136.362
Occupation 12	317.620
Occupation 13	128.270
Occupation 14	580.101
Occupation 15	224.538
Occupation 16	368.201
Occupation 17	374.962
Occupation 18	23.172
Occupation 19	170.072
Occupation 2	725.695
Occupation 20	734.290
Occupation 3	717.076
Occupation 4	1522.643
Occupation 5	195.950
Occupation 6	740.798
Occupation 7	911.776
Occupation 8	33.795
Occupation 9	502.065

Name: Purchase, dtype: float64

spending_of_men_across_occupation:

Occupation_category	
Occupation 0	4755.231
Occupation 1	2718.074
Occupation 10	830.409
Occupation 11	931.154
Occupation 12	2736.874

Occupation 13 590.925
Occupation 14 2014.446
Occupation 15 965.064
Occupation 16 2015.268
Occupation 17 3557.853
Occupation 18 584.043
Occupation 19 566.935
Occupation 2 1654.591
Occupation 20 2231.415
Occupation 3 902.945
Occupation 4 5139.802
Occupation 5 940.547
Occupation 6 1143.370
Occupation 7 4661.940
Occupation 8 113.579
Occupation 9 41.336
Name: Purchase, dtype: float64

8. How does the distribution of married and single customers vary across different occupation categories?

ANS(CODE):

```
import pandas as pd
import matplotlib.pyplot as plt

data = pd.read_csv('walmart_data.csv')
married = data[data['Marital_category'] == 'Married']
single = data[data['Marital_category'] == 'Single']
married_occupation = married.groupby('Occupation_category')['Marital_category'].count()
single_occupation = single.groupby('Occupation_category')['Marital_category'].count()
x = range(len(married_occupation))
width = 0.4
plt.bar([i - width/2 for i in x], married_occupation.values, width=width, label='Married')
```

```

plt.bar([i + width/2 for i in x], single_occupation.values, width=width, label='Single')

plt.xlabel('Occupation')

plt.ylabel('Number of Customers')

plt.title('Married vs Single Customers Across Occupations')

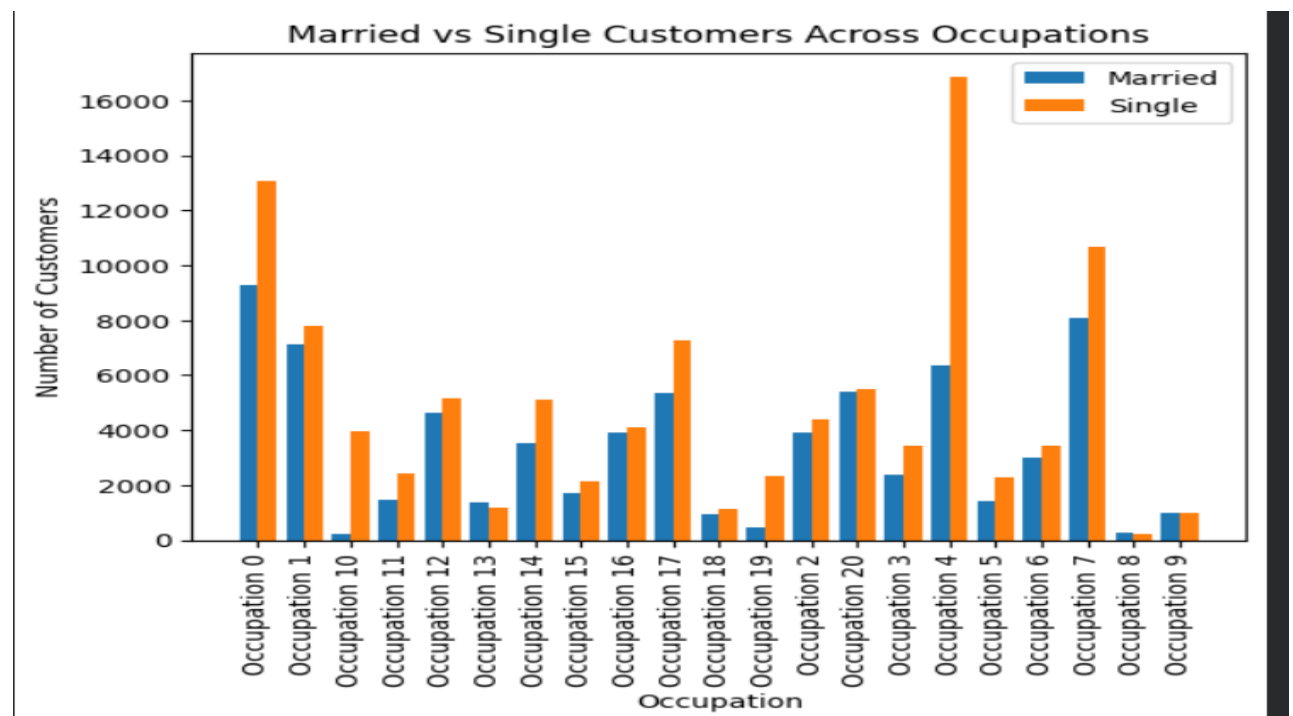
plt.xticks(x, married_occupation.index, rotation=90)

plt.legend()

plt.show()

```

OUTPUT:



9. How does the distribution of married and single customers vary across different city categories?

ANS(CODE):

```

import pandas as pd

data = pd.read_csv('walmart_data.csv')

married = data[data['Marital_category'] == 'Married']

single = data[data['Marital_category'] == 'Single']

marriage_per_city = married.groupby('City_Category')['User_ID'].count()

singles_per_city = single.groupby('City_Category')['User_ID'].count()

```

```

x = range(len(marriage_per_city))

width = 0.4

plt.bar([i - width/2 for i in x], marriage_per_city.values, width=width, label='Married')

plt.bar([i + width/2 for i in x], singles_per_city.values, width=width, label='Single')

plt.xlabel('City-categories')

plt.ylabel('Number of Customers')

plt.title('Married vs Single Customers Across three city categories')

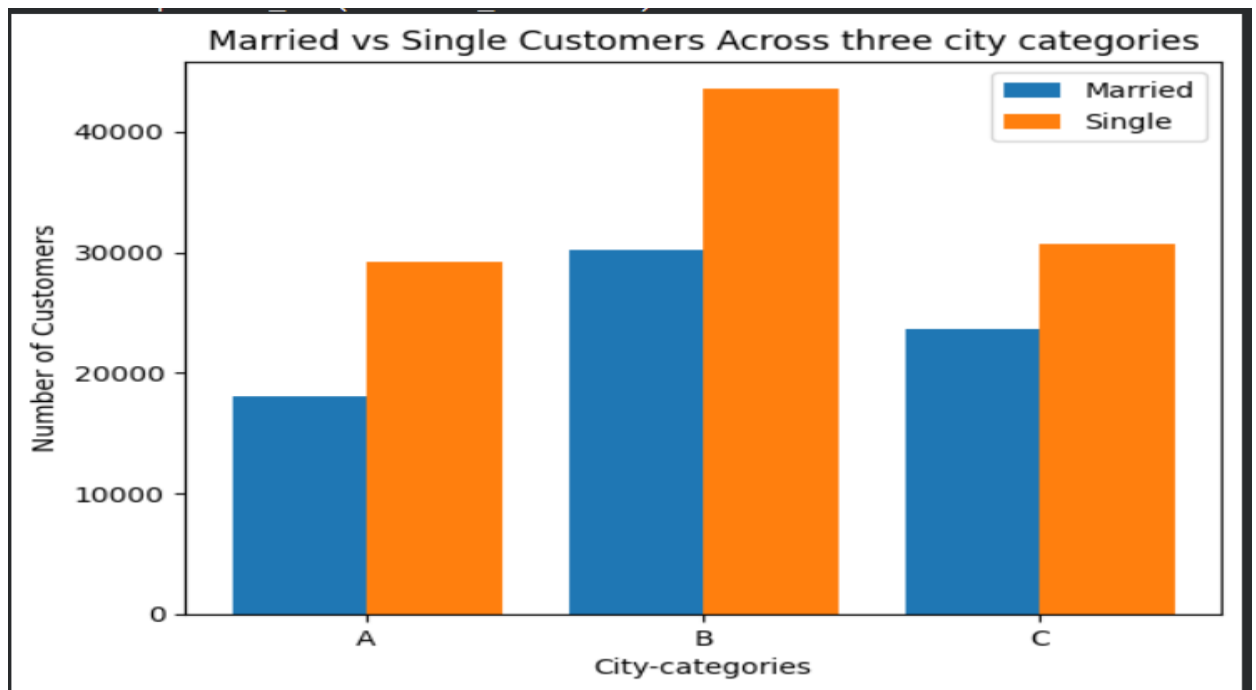
plt.xticks(x, marriage_per_city.index)

plt.legend()

plt.show()

```

OUTPUT:



10. Does customer purchase amount significantly differ across occupation categories and years of stay in the current city, and is there an interaction effect between occupation and years of stay?

HYPOTHESIS:

1. Occupation Category

- **H₀:** There is no significant difference in purchase amount among different occupation categories.
- **H₁:** There is a significant difference in purchase amount among different occupation categories.

2. Stay in Current City

- H_0 : There is no significant difference in purchase amount among customers based on their years of stay in the current city.
- H_1 : There is a significant difference in purchase amount among customers based on their years of stay in the current city.

3. Interaction Effect

- H_0 : There is no significant interaction effect between occupation category and years of stay in the current city on purchase amount.
- H_1 : There is a significant interaction effect between occupation category and years of stay in the current city on purchase amount.

ANS(CODE):

```
import pandas as pd

from statsmodels.formula.api import ols

import statsmodels.api as sm

data = pd.read_csv('walmart_data.csv')

sub_chart = data[['Occupation_category', 'Stay_In_Current_City_Years', 'Purchase']]

test = ols('Purchase ~ C(Occupation_category) * C(Stay_In_Current_City_Years)',
data=sub_chart).fit()

annova_table = sm.stats.anova_lm(test, typ=2)

print(annova_table)
```

OUTPUT:

	sum_sq	df	F	PR(>F)
C(Occupation_category)	1.786820e+10	20.0	36.262278	1.474102e-133
C(Stay_In_Current_City_Years)	2.868237e+08	4.0	2.910445	3.308754e-02
C(Occupation_category):C(Stay_In_Current_City_Years)	1.022358e+10	80.0	5.187014	4.547122e-46
Residual	4.316406e+12	175197.0	NaN	NaN

11. Does customer purchase amount significantly differ based on gender and product category, and is there a significant interaction effect between gender and product category on purchase amount?

Gender

H_0 (Null Hypothesis): There is no significant difference in purchase amount between different genders.

H_1 (Alternative Hypothesis): There is a significant difference in purchase amount between different genders.

Product Category

H₀ (Null Hypothesis): There is no significant difference in purchase amount among different product categories.

H₁ (Alternative Hypothesis): There is a significant difference in purchase amount among different product categories.

Interaction Effect — Gender × Product Category

H₀ (Null Hypothesis): There is no significant interaction effect between gender and product category on purchase amount.

H₁ (Alternative Hypothesis): There is a significant interaction effect between gender and product category on purchase amount.

ANS(CODE):

```
import pandas as pd

from statsmodels.formula.api import ols

import statsmodels.api as sm

data = pd.read_csv('walmart_data.csv')

sub_chart = data[['Gender', 'Product_Category', 'Purchase']]

test = ols('Purchase ~ C(Gender) * C(Product_Category)', data=sub_chart).fit()

annova_table = sm.stats.anova_lm(test, typ=2)

print(annova_table)
```

OUTPUT:

	sum_sq	df	F	PR(>F)
C(Gender)	1.592100e+08	1.0	17.365479	3.084234e-05
C(Product_Category)	8.784168e+12	19.0	50427.038798	0.000000e+00
C(Gender):C(Product_Category)	1.392372e+09	19.0	7.993151	9.624629e-23
Residual	5.042761e+12	550028.0	NaN	NaN

12. Using the Central Limit Theorem, how do the sampling distributions and 95% confidence intervals of mean purchase amounts compare between male and female customers?

ANS(CODE):

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np
```

```
data = pd.read_csv('walmart_data.csv')

male = data[data['Gender'] == 'M']['Purchase']
female = data[data['Gender'] == 'F']['Purchase']

#means of condition
male_mean = male.mean()
female_mean = female.mean()

#empty array for storage
male_sample_means = []
female_sample_means = []

#std for female and male
male_std = male.std()
female_std = female.std()

for i in range(1000):

    male_sample = male.sample(100, replace=True)

    male_sample_means.append(male_sample.mean())

    female_sample = female.sample(100, replace=True)

    female_sample_means.append(female_sample.mean())

# Confidence Interval
n = 100
z = 1.96

# Male Confidence Interval
```

```

male_sample_mean = np.mean(male_sample_means)
male_margin_of_error = z * (male_std / np.sqrt(n))
male_lower = male_sample_mean - male_margin_of_error
male_upper = male_sample_mean + male_margin_of_error

# Female Confidence Interval
female_sample_mean = np.mean(female_sample_means)
female_margin_of_error = z * (female_std / np.sqrt(n))
female_lower = female_sample_mean - female_margin_of_error
female_upper = female_sample_mean + female_margin_of_error

#printing the output
print("Male Sample Mean:", male_sample_mean)
print("Male 95% Confidence Interval:", male_lower, male_upper)
print("Female Sample Mean:", female_sample_mean)
print("Female 95% Confidence Interval:", female_lower, female_upper)

#plotting the graph
sns.histplot(male_sample_means, kde=True, label='Male')
sns.histplot(female_sample_means, kde=True, label='Female')
plt.title("Distribution of Sample Means of Purchase by Gender")
plt.xlabel("Sample Mean Purchase")
plt.ylabel("Frequency")
plt.legend()
plt.show()

```

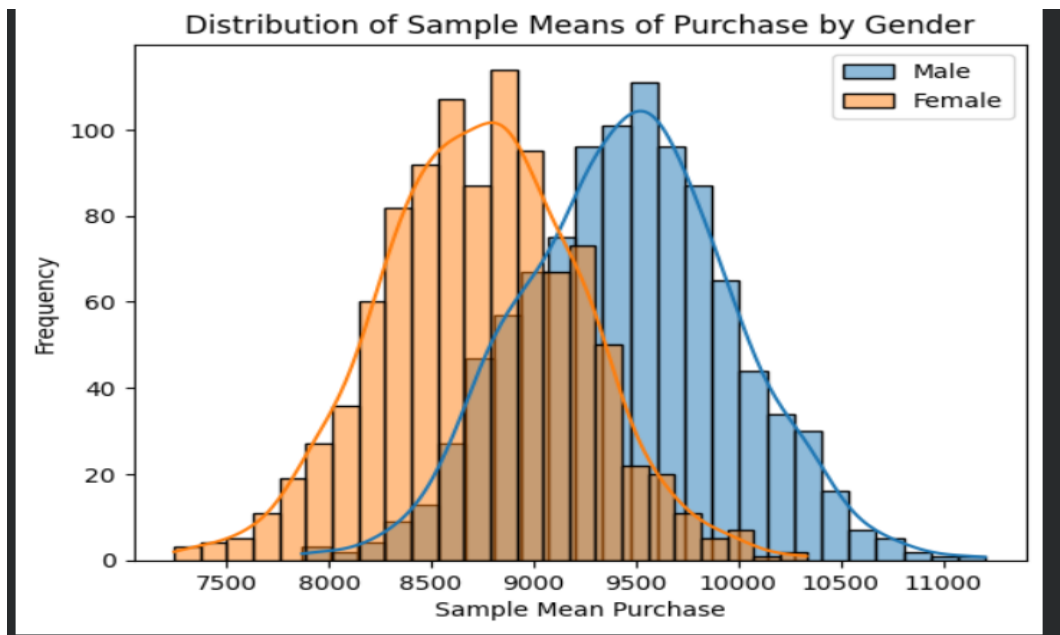
OUTPUT:

Male Sample Mean: 9457.67086

Male 95% Confidence Interval: 8459.602362883523 10455.739357116478

Female Sample Mean: 8729.52091

Female 95% Confidence Interval: 7795.140123887041 9663.901696112958



13. Using the Central Limit Theorem, how does the average purchase amount differ between married and single customers based on their sampling distributions and 95% confidence intervals?

ANS(CODE):

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

import numpy as np

data = pd.read_csv('walmart_data.csv')

married = data[data['Marital_Status'] == 1]['Purchase']

single = data[data['Marital_Status'] == 0]['Purchase']

# means of condition

married_mean = married.mean()

single_mean = single.mean()

# empty array for storage

married_sample_means = []

single_sample_means = []

# std for married and single

married_std = married.std()

single_std = single.std()
```

```

for i in range(1000):

    married_sample = married.sample(100, replace=True)

    married_sample_means.append(married_sample.mean())

    single_sample = single.sample(100, replace=True)

    single_sample_means.append(single_sample.mean())
# Confidence Interval
n = 100
z = 1.96
# Married Confidence Interval
married_sample_mean = np.mean(married_sample_means)
married_margin_of_error = z * (married_std / np.sqrt(n))
married_lower = married_sample_mean - married_margin_of_error
married_upper = married_sample_mean + married_margin_of_error
# Single Confidence Interval
single_sample_mean = np.mean(single_sample_means)
single_margin_of_error = z * (single_std / np.sqrt(n))
single_lower = single_sample_mean - single_margin_of_error
single_upper = single_sample_mean + single_margin_of_error
# printing the output
print("Married Sample Mean:", round(married_sample_mean,2))
print("Married 95% Confidence Interval:", married_lower, married_upper)
print("Single Sample Mean:", round(single_sample_mean,2))
print("Single 95% Confidence Interval:", single_lower, single_upper)
# plotting the graph
sns.histplot(married_sample_means, kde=True, label='Married')
sns.histplot(single_sample_means, kde=True, label='Single')
plt.title("Distribution of Sample Means of Purchase by Marital Status")

```

```
plt.xlabel("Sample Mean Purchase")
plt.ylabel("Frequency")
plt.legend()
plt.show()
```

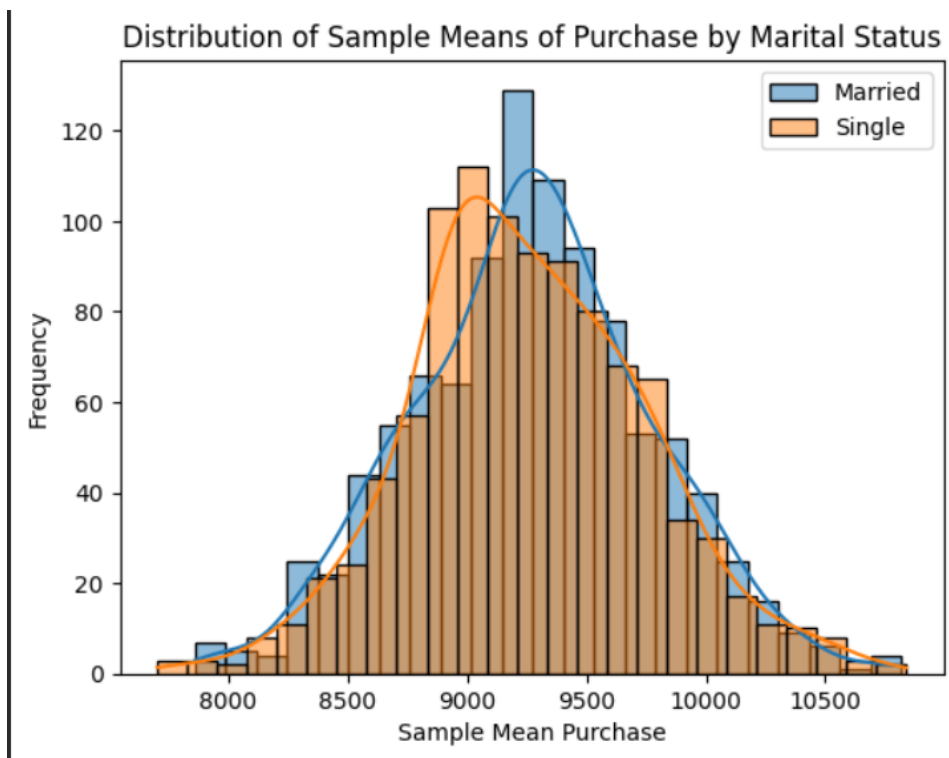
OUTPUT:

Married Sample Mean: 9268.54

Married 95% Confidence Interval: 8285.229383952546 10251.853156047453

Single Sample Mean: 9256.2

Single 95% Confidence Interval: 8270.843729699807 10241.564090300193



14. Using the Central Limit Theorem, how does the average purchase amount differ across different age groups based on their sampling distributions and 95% confidence intervals?

ANS(CODE):

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import numpy as np
```

```
data = pd.read_csv('walmart_data.csv')
```

```
# Creating age groups
```

```
data['Age_Group'] = data['Age'].replace({  
    '0-17': '0-17',  
    '18-25': '18-25',  
    '26-35': '26-35',  
    '36-45': '36-50',  
    '46-50': '36-50',  
    '51-55': '51+',  
    '55+': '51+'  
})
```

```
# Separate purchase amounts by age group
```

```
age_0_17 = data[data['Age_Group'] == '0-17']['Purchase']  
age_18_25 = data[data['Age_Group'] == '18-25']['Purchase']  
age_26_35 = data[data['Age_Group'] == '26-35']['Purchase']  
age_36_50 = data[data['Age_Group'] == '36-50']['Purchase']  
age_51_plus = data[data['Age_Group'] == '51+']['Purchase']
```

```
# Empty arrays for storage
```

```
age_0_17_sample_means = []  
age_18_25_sample_means = []  
age_26_35_sample_means = []  
age_36_50_sample_means = []  
age_51_plus_sample_means = []
```

```
# Standard deviation
```

```
age_0_17_std = age_0_17.std()  
age_18_25_std = age_18_25.std()  
age_26_35_std = age_26_35.std()
```

```

age_36_50_std = age_36_50.std()
age_51_plus_std = age_51_plus.std()

# CLT sampling
for i in range(1000):

    age_0_17_sample = age_0_17.sample(100, replace=True)
    age_0_17_sample_means.append(age_0_17_sample.mean())

    age_18_25_sample = age_18_25.sample(100, replace=True)
    age_18_25_sample_means.append(age_18_25_sample.mean())

    age_26_35_sample = age_26_35.sample(100, replace=True)
    age_26_35_sample_means.append(age_26_35_sample.mean())

    age_36_50_sample = age_36_50.sample(100, replace=True)
    age_36_50_sample_means.append(age_36_50_sample.mean())

    age_51_plus_sample = age_51_plus.sample(100, replace=True)
    age_51_plus_sample_means.append(age_51_plus_sample.mean())

# Confidence Interval

n = 100
z = 1.96

# 0-17 Confidence Interval
age_0_17_sample_mean = np.mean(age_0_17_sample_means)
age_0_17_margin = z * (age_0_17_std / np.sqrt(n))
age_0_17_lower = age_0_17_sample_mean - age_0_17_margin
age_0_17_upper = age_0_17_sample_mean + age_0_17_margin

```

```
# 18-25 Confidence Interval
```

```
age_18_25_sample_mean = np.mean(age_18_25_sample_means)
```

```
age_18_25_margin = z * (age_18_25_std / np.sqrt(n))
```

```
age_18_25_lower = age_18_25_sample_mean - age_18_25_margin
```

```
age_18_25_upper = age_18_25_sample_mean + age_18_25_margin
```

```
# 26-35 Confidence Interval
```

```
age_26_35_sample_mean = np.mean(age_26_35_sample_means)
```

```
age_26_35_margin = z * (age_26_35_std / np.sqrt(n))
```

```
age_26_35_lower = age_26_35_sample_mean - age_26_35_margin
```

```
age_26_35_upper = age_26_35_sample_mean + age_26_35_margin
```

```
# 36-50 Confidence Interval
```

```
age_36_50_sample_mean = np.mean(age_36_50_sample_means)
```

```
age_36_50_margin = z * (age_36_50_std / np.sqrt(n))
```

```
age_36_50_lower = age_36_50_sample_mean - age_36_50_margin
```

```
age_36_50_upper = age_36_50_sample_mean + age_36_50_margin
```

```
# 51+ Confidence Interval
```

```
age_51_plus_sample_mean = np.mean(age_51_plus_sample_means)
```

```
age_51_plus_margin = z * (age_51_plus_std / np.sqrt(n))
```

```
age_51_plus_lower = age_51_plus_sample_mean - age_51_plus_margin
```

```
age_51_plus_upper = age_51_plus_sample_mean + age_51_plus_margin
```

```
# Printing the output
```

```
print("0-17 Sample Mean:", age_0_17_sample_mean)
```

```
print("0-17 95% Confidence Interval:", age_0_17_lower, age_0_17_upper)
```

```
print("18-25 Sample Mean:", age_18_25_sample_mean)
```

```
print("18-25 95% Confidence Interval:", age_18_25_lower, age_18_25_upper)
```

```
print("26-35 Sample Mean:", age_26_35_sample_mean)
```

```
print("26-35 95% Confidence Interval:", age_26_35_lower, age_26_35_upper)
```

```

print("36-50 Sample Mean:", age_36_50_sample_mean)
print("36-50 95% Confidence Interval:", age_36_50_lower, age_36_50_upper)
print("51+ Sample Mean:", age_51_plus_sample_mean)
print("51+ 95% Confidence Interval:", age_51_plus_lower, age_51_plus_upper)

# Plotting the graph
sns.histplot(age_0_17_sample_means, kde=True, label='0-17')
sns.histplot(age_18_25_sample_means, kde=True, label='18-25')
sns.histplot(age_26_35_sample_means, kde=True, label='26-35')
sns.histplot(age_36_50_sample_means, kde=True, label='36-50')
sns.histplot(age_51_plus_sample_means, kde=True, label='51+')
plt.title("Distribution of Sample Means of Purchase by Age Group")
plt.xlabel("Sample Mean Purchase")
plt.ylabel("Frequency")
plt.legend()
plt.show()

```

OUTPUT:

0-17 Sample Mean: 8926.37503

0-17 95% Confidence Interval: 7924.596676983456 9928.153383016543

18-25 Sample Mean: 9175.829699999998

18-25 95% Confidence Interval: 8189.102588553389 10162.556811446608

26-35 Sample Mean: 9250.14481

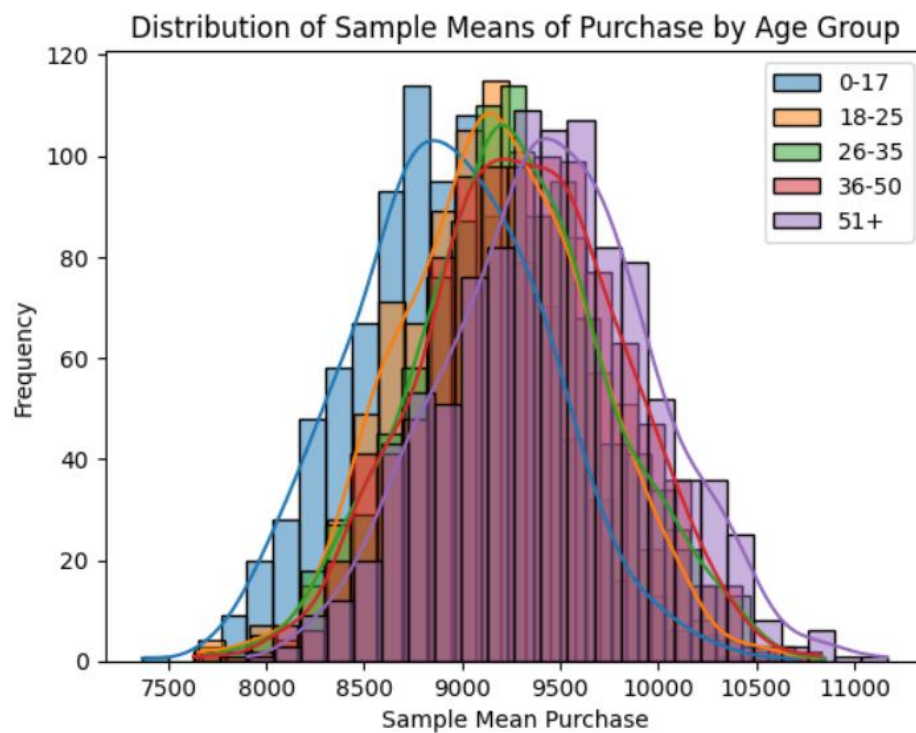
26-35 95% Confidence Interval: 8268.081458611427 10232.208161388573

36-50 Sample Mean: 9303.24207

36-50 95% Confidence Interval: 8321.88294258175 10284.601197418251

51+ Sample Mean: 9452.21858

51+ 95% Confidence Interval: 8460.230930679481 10444.20622932052

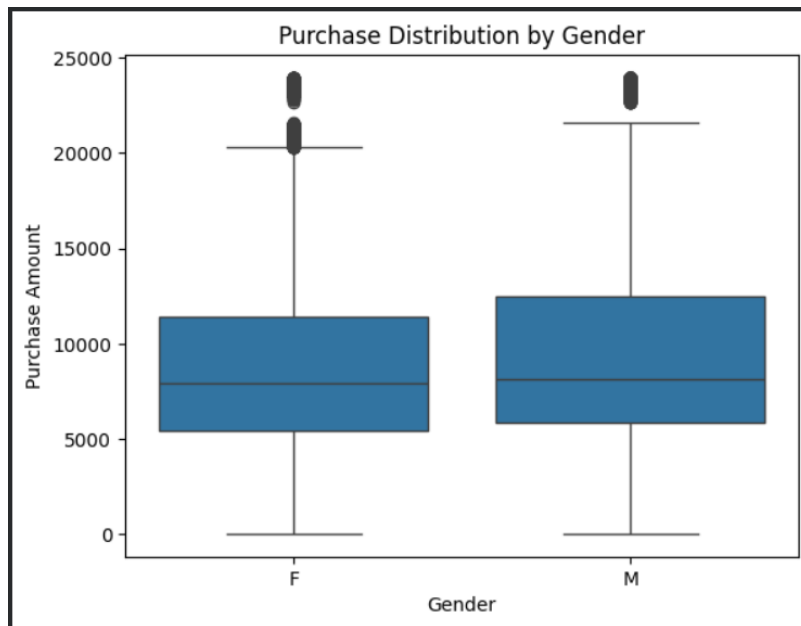


15. How does the distribution of purchase amounts vary between male and female customers?

ANS(CODE):

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
data = pd.read_csv('walmart_data.csv')
sns.boxplot(x='Gender', y='Purchase', data=data)
plt.title('Purchase Distribution by Gender')
plt.xlabel('Gender')
plt.ylabel('Purchase Amount')
plt.show()
```

OUTPUT:



16. Using the Central Limit Theorem, how do the sample means and 95% confidence intervals compare across gender, marital status, stay in current city, and different age groups in the Walmart dataset?"

ANS(CODE):

```
import pandas as pd
import numpy as np

data = pd.read_csv('walmart_data.csv')
male = data[data['Gender'] == 'M']['Purchase']
female = data[data['Gender'] == 'F']['Purchase']

male_sample_means = []
female_sample_means = []

male_std = male.std()
female_std = female.std()

for i in range(1000):

    male_sample = male.sample(100, replace=True)
    male_sample_means.append(male_sample.mean())

    female_sample = female.sample(100, replace=True)
    female_sample_means.append(female_sample.mean())
```

```

data['Stay_In_Current_City_Years'] = data[
    'Stay_In_Current_City_Years'
].replace('4+', '4').astype(int)

male_city = data[data['Gender'] == 'M']['Stay_In_Current_City_Years']
female_city = data[data['Gender'] == 'F']['Stay_In_Current_City_Years']

male_city_sample_means = []
female_city_sample_means = []

male_city_std = male_city.std()
female_city_std = female_city.std()

for i in range(1000):

    male_city_sample = male_city.sample(100, replace=True)
    male_city_sample_means.append(male_city_sample.mean())

    female_city_sample = female_city.sample(100, replace=True)
    female_city_sample_means.append(female_city_sample.mean())

# =====
# 3. MARITAL STATUS - PURCHASE
# =====

married = data[data['Marital_Status'] == 1]['Purchase']
single = data[data['Marital_Status'] == 0]['Purchase']

married_sample_means = []
single_sample_means = []

```

```
married_std = married.std()
```

```
single_std = single.std()
```

```
for i in range(1000):
```

```
    married_sample = married.sample(100, replace=True)
```

```
    married_sample_means.append(married_sample.mean())
```

```
    single_sample = single.sample(100, replace=True)
```

```
    single_sample_means.append(single_sample.mean())
```

```
data['Age_Group'] = data['Age'].replace({
```

```
    '0-17': '0-17',
```

```
    '18-25': '18-25',
```

```
    '26-35': '26-35',
```

```
    '36-45': '36-50',
```

```
    '46-50': '36-50',
```

```
    '51-55': '51+',
```

```
    '55+': '51+'
```

```
})
```

```
age_0_17 = data[data['Age_Group'] == '0-17']['Purchase']
```

```
age_18_25 = data[data['Age_Group'] == '18-25']['Purchase']
```

```
age_26_35 = data[data['Age_Group'] == '26-35']['Purchase']
```

```
age_36_50 = data[data['Age_Group'] == '36-50']['Purchase']
```

```
age_51_plus = data[data['Age_Group'] == '51+']['Purchase']
```

```
age_0_17_sample_means = []
```

```
age_18_25_sample_means = []
```

```
age_26_35_sample_means = []
```

```
age_36_50_sample_means = []
```

```
age_51_plus_sample_means = []
```

```

age_0_17_std = age_0_17.std()
age_18_25_std = age_18_25.std()
age_26_35_std = age_26_35.std()
age_36_50_std = age_36_50.std()
age_51_plus_std = age_51_plus.std()

for i in range(1000):

    age_0_17_sample = age_0_17.sample(100, replace=True)
    age_0_17_sample_means.append(age_0_17_sample.mean())

    age_18_25_sample = age_18_25.sample(100, replace=True)
    age_18_25_sample_means.append(age_18_25_sample.mean())

    age_26_35_sample = age_26_35.sample(100, replace=True)
    age_26_35_sample_means.append(age_26_35_sample.mean())

    age_36_50_sample = age_36_50.sample(100, replace=True)
    age_36_50_sample_means.append(age_36_50_sample.mean())

    age_51_plus_sample = age_51_plus.sample(100, replace=True)
    age_51_plus_sample_means.append(age_51_plus_sample.mean())

# =====
# 5. COMBINE ALL CLT GROUPS
# =====

groups = {

    # Gender - Purchase

```

```

'Male Purchase': (male_sample_means, male_std),
'Female Purchase': (female_sample_means, female_std),

# Gender - Stay in Current City
'Male Stay in City': (male_city_sample_means, male_city_std),
'Female Stay in City': (female_city_sample_means, female_city_std),

# Marital Status - Purchase
'Married Purchase': (married_sample_means, married_std),
'Single Purchase': (single_sample_means, single_std),

# Age - Purchase
'Age 0-17 Purchase': (age_0_17_sample_means, age_0_17_std),
'Age 18-25 Purchase': (age_18_25_sample_means, age_18_25_std),
'Age 26-35 Purchase': (age_26_35_sample_means, age_26_35_std),
'Age 36-50 Purchase': (age_36_50_sample_means, age_36_50_std),
'Age 51+ Purchase': (age_51_plus_sample_means, age_51_plus_std)
}

# =====

# 6. CONFIDENCE INTERVALS

# =====

confidence_levels = {
    90: 1.645,
    95: 1.96,
    99: 2.576
}

n = 100

```

```

confidence_intervals = []

for group, (sample_means, std) in groups.items():

    sample_mean = np.mean(sample_means)

    for confidence, z in confidence_levels.items():

        margin = z * (std / np.sqrt(n))

        lower = sample_mean - margin
        upper = sample_mean + margin

        confidence_intervals.append([
            group,
            confidence,
            sample_mean,
            lower,
            upper
        ])

# =====
# 7. FINAL CONFIDENCE INTERVAL TABLE
# =====

ci_table = pd.DataFrame(
    confidence_intervals,
    columns=[
        'CLT Group',
        'Confidence Level',
        'Sample Mean',

```

```

        'Lower Limit',
        'Upper Limit'
    ]
)

# Round values for a cleaner table
ci_table['Sample Mean'] = ci_table['Sample Mean'].round(2)
ci_table['Lower Limit'] = ci_table['Lower Limit'].round(2)
ci_table['Upper Limit'] = ci_table['Upper Limit'].round(2)
ci_table['Confidence Level'] = (
    ci_table['Confidence Level'].astype(str) + '%'
)

print("\nALL CLT CONFIDENCE INTERVALS")
display(ci_table)

# =====
# 8. 95% CONFIDENCE INTERVALS ONLY
# =====

ci_95 = ci_table[
    ci_table['Confidence Level'] == '95%'
].reset_index(drop=True)
print("\n95% CONFIDENCE INTERVALS")
display(ci_95)

```

OUTPUT:

ALL CLT CONFIDENCE INTERVALS

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
0	Male Purchase	90%	9434.00	8596.33	10271.66
1	Male Purchase	95%	9434.00	8435.93	10432.07

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
2	Male Purchase	99%	9434.00	8122.25	10745.74
3	Female Purchase	90%	8732.31	7948.10	9516.52
4	Female Purchase	95%	8732.31	7797.93	9666.69
5	Female Purchase	99%	8732.31	7504.27	9960.35
6	Male Stay in City	90%	1.87	1.66	2.08
7	Male Stay in City	95%	1.87	1.61	2.12
8	Male Stay in City	99%	1.87	1.53	2.20
9	Female Stay in City	90%	1.83	1.62	2.03
10	Female Stay in City	95%	1.83	1.58	2.07
11	Female Stay in City	99%	1.83	1.50	2.15
12	Married Purchase	90%	9264.03	8438.75	10089.31
13	Married Purchase	95%	9264.03	8280.72	10247.34
14	Married Purchase	99%	9264.03	7971.68	10556.39

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
15	Single Purchase	90%	9267.33	8440.33	10094.32
16	Single Purchase	95%	9267.33	8281.97	10252.69
17	Single Purchase	99%	9267.33	7972.28	10562.37
18	Age 0-17 Purchase	90%	8933.95	8093.17	9774.73
19	Age 0-17 Purchase	95%	8933.95	7932.17	9935.73
20	Age 0-17 Purchase	99%	8933.95	7617.33	10250.57
21	Age 18-25 Purchase	90%	9153.97	8325.83	9982.12
22	Age 18-25 Purchase	95%	9153.97	8167.25	10140.70
23	Age 18-25 Purchase	99%	9153.97	7857.13	10450.82
24	Age 26-35 Purchase	90%	9251.91	8427.67	10076.14
25	Age 26-35 Purchase	95%	9251.91	8269.84	10233.97
26	Age 26-35 Purchase	99%	9251.91	7961.19	10542.62
27	Age 36-50 Purchase	90%	9280.21	8456.57	10103.85

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
28	Age 36-50 Purchase	95%	9280.21	8298.85	10261.57
29	Age 36-50 Purchase	99%	9280.21	7990.42	10569.99
30	Age 51+ Purchase	90%	9478.16	8645.60	10310.72
31	Age 51+ Purchase	95%	9478.16	8486.17	10470.15
32	Age 51+ Purchase	99%	9478.16	8174.41	10781.92

95% CONFIDENCE INTERVALS

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
0	Male Purchase	95%	9434.00	8435.93	10432.07
1	Female Purchase	95%	8732.31	7797.93	9666.69
2	Male Stay in City	95%	1.87	1.61	2.12
3	Female Stay in City	95%	1.83	1.58	2.07
4	Married Purchase	95%	9264.03	8280.72	10247.34
5	Single Purchase	95%	9267.33	8281.97	10252.69

	CLT Group	Confidence Level	Sample Mean	Lower Limit	Upper Limit
6	Age 0-17 Purchase	95%	8933.95	7932.17	9935.73
7	Age 18-25 Purchase	95%	9153.97	8167.25	10140.70
8	Age 26-35 Purchase	95%	9251.91	8269.84	10233.97
9	Age 36-50 Purchase	95%	9280.21	8298.85	10261.57
10	Age 51+ Purchase	95%	9478.16	8486.17	10470.15

17. How do the mean and median purchase amounts compare, and what does the outlier analysis reveal about the distribution of customer purchase amounts in the Walmart dataset?

ANS(CODE):

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
data = pd.read_csv('walmart_data.csv')
print("Statistical Summary of Purchase Amount:")
print(data['Purchase'].describe())
mean_purchase = data['Purchase'].mean()
median_purchase = data['Purchase'].median()
print("\nMean Purchase Amount:", mean_purchase)
print("Median Purchase Amount:", median_purchase)
print("\nMean vs Median:")
if mean_purchase > median_purchase:
    print("The mean is higher than the median, indicating that higher purchase values may be influencing the average.")
```

```

elif mean_purchase < median_purchase:

    print("The mean is lower than the median, indicating that lower purchase values may be
    influencing the average.")

else:

    print("The mean and median are equal.")

Q1 = data['Purchase'].quantile(0.25)
Q3 = data['Purchase'].quantile(0.75)
IQR = Q3 - Q1

print("\nFirst Quartile (Q1):", Q1)
print("Third Quartile (Q3):", Q3)
print("Interquartile Range (IQR):", IQR)

lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR

print("\nLower Outlier Limit:", lower_limit)
print("Upper Outlier Limit:", upper_limit)

outliers = data[

    (data['Purchase'] < lower_limit) |

    (data['Purchase'] > upper_limit)

]

print("\nNumber of Outliers:", len(outliers))
print("\nOutlier Purchase Values:")
print(outliers['Purchase'].head(10))

plt.figure(figsize=(8, 5))
plt.boxplot(data['Purchase'])
plt.title('Boxplot of Purchase Amount')
plt.ylabel('Purchase Amount')
plt.show()

```

OUTPUT:

Statistical Summary of Purchase Amount:

count 250445.000000

mean 9313.668754

std **4967.982450**
min **185.000000**
25% **5862.000000**
50% **8058.000000**
75% **12055.000000**
max **23961.000000**

Name: Purchase, dtype: float64

Mean Purchase Amount: 9313.668753618558

Median Purchase Amount: 8058.0

Mean vs Median:

The mean is higher than the median, indicating that higher purchase values may be influencing the average.

First Quartile (Q1): 5862.0

Third Quartile (Q3): 12055.0

Interquartile Range (IQR): 6193.0

Lower Outlier Limit: -3427.5

Upper Outlier Limit: 21344.5

Number of Outliers: 1237

Outlier Purchase Values:

343 23603.0

375 23792.0

652 23233.0

736 23595.0

1041 23341.0

1445 23826.0

1902 23139.0

3166 23159.0

3167 23741.0

3172 23728.0

Name: Purchase, dtype: float64



INSIGHTS:

- Based on the **first and second questions**, we obtained some basic insights about the dataset. The original dataset contains 9 columns with 550,068 entries. The average purchase amount across all genders, cities, and occupations is ₹9,263.969, with the maximum purchase amount being ₹23,961 and the minimum purchase amount being ₹12.
- There were some missing values in almost all columns except user_id and product_id, which needed to be removed before further analysis. There were also some columns that needed to be added based on the existing data in the main table to make the dataset easier to analyze. Therefore, the Occupation_category and Marital_category columns were added to the dataset. The null values were removed, and the upgraded dataset was saved for further analysis.
- Based on **Question 3**, ****City C has the highest number of users, with 3,139 users****. Among the product categories, ****Category 8 has the highest number of different products, with 1,047 products****, while ****Category 19 has the lowest, with only 2 products****. In terms of total spending, ****Product Category 1 has the highest spending at 19,100.138****, whereas ****Product Category 19 has the lowest spending at 0.594****. This indicates that Product Category 1 contributes significantly more to overall sales, while Category 19 has very limited product variety and spending.
- Based on **Question 4**, the highest total spending among occupations is from ****Occupation 4 (₹6,662.445)****, whereas the lowest total spending is from ****Occupation 8 (₹147.374)****. It is also observed that ****Single customers have the highest total spending (₹30,089.274)****, while ****Married customers have the lowest**

total spending (₹20,868.853)**. Furthermore, customers who have stayed in their current city for **1 year have the highest total spending (₹17,968.725)**, whereas customers who have stayed for **0 years have the lowest total spending (₹6,829.792)**.

- Based on **Question 5**, we get the insight that the total number of male users (414,259) is higher than the number of female users (135,809). Among unmarried females, the highest count is in the 26–35 age group (30,078), while the lowest count is in the 55+ age group (1,908). Among married females, the highest count is also in the 26–35 age group (20,674), whereas the lowest count is in the 55+ age group (3,175).
- Similarly, among unmarried males, the highest count is in the 26–35 age group (103,218), while the lowest count is in the 55+ age group (5,975). Among married males, the highest count is in the 26–35 age group (65,617), whereas the lowest count is in the 18–25 age group (14,845).
- In terms of spending, unmarried female customers have the highest total spending (6,841.541), compared with married female customers (5,020.785). Similarly, unmarried male customers have the highest total spending (23,247.733), while married male customers have lower total spending (15,848.068). Overall, unmarried customers have higher total spending than married customers for both genders.
- Based on **Question 6**, City B has the highest total spending of 21,155.336, followed by City C with 16,638.075, while City A has the lowest total spending of 13,164.717. Product Category 1 has the highest spending in all three city categories, whereas Product Category 19 has the lowest spending in all three cities.
- In terms of years stayed in the current city, customers who have stayed for 1 year have the highest total spending of 17,928.725, while customers who have stayed for 0 years have the lowest spending of 6,829.792. For age groups, customers aged 26–35 have the highest total spending of 20,317.706, whereas the 0–17 age group has the lowest spending of 1,349.132.
- The average spending per customer is highest in City A at 1,259,781.494, followed by City B at 1,239,328.415, while City C has the lowest average spending per customer at 530,043.796. Overall, the results show that spending varies across cities, age groups, and years of residence, with City B generating the highest total spending and customers aged 26–35 contributing the highest spending among the age groups.
- Based on **Question 7**, the occupation with the highest overall spending is Occupation 4 (6,662.445), whereas the occupation with the lowest overall spending is Occupation 8 (147.374). Among female customers, Occupation 0 has the highest spending (1,598.838), while Occupation 18 has the lowest spending (23.172). Among male customers, Occupation 4 has the highest spending (5,139.802), whereas Occupation 9 has the lowest spending (41.336).
- Based on **Question 8**, Occupation 4 has the highest number of Single customers, whereas Occupation 0 has the highest number of Married customers.
- Based on **Question 9**, the highest number of Married customers is in City B, whereas City A has the lowest number of Married customers. Similarly, City B has the highest number of Single customers, while the numbers of Single customers in City A and City C are very similar, making the difference between them relatively small.
- Based on **Question 10**, There is a significant interaction effect between occupation category and years of stay in the current city on customer purchase amount. This

means that the effect of occupation on purchase amount depends on the customer's years of stay in the current city.

- Based on **Question 11**, there is a significant interaction effect between gender and product category on customer purchase amount. This indicates that the effect of product category on purchase amount differs between male and female customers.
- Based on **Question 12**, using the Central Limit Theorem, the sample mean purchase amount for male customers (₹9,457.67) is higher than that of female customers (₹8,729.52). The 95% confidence interval for male customers is ₹8,459.60 to ₹10,455.74, while for female customers it is ₹7,795.14 to ₹9,663.90. The confidence intervals overlap, indicating that there is some similarity in the estimated average purchase amounts between male and female customers. However, the sample results show that male customers have a higher average purchase amount than female customers.
- According to **Question 13**, married and single customers have almost the same average purchase amount. Married customers have a slightly higher average purchase amount of ₹9,268.54, compared to ₹9,256.20 for single customers. The 95% confidence intervals for married customers (₹8,285.23–₹10,251.85) and single customers (₹8,270.84–₹10,241.56) also overlap considerably. This shows that there is very little difference in the average purchase amount between married and single customers.
- Based on **Question 14**, the average purchase amount shows an increasing trend across the age groups. Customers aged 0–17 have the lowest average purchase amount of ₹8,926.38, while customers aged 51+ have the highest average purchase amount of ₹9,452.22. The average purchase amount gradually increases across the 18–25, 26–35, and 36–50 age groups. However, the 95% confidence intervals of all age groups overlap considerably, showing that the differences in average purchase amounts between the age groups are relatively small based on the sampling results.
- Based on **Question 15**, the distribution of purchase amounts varies between male and female customers. The boxplot shows that male customers generally have a higher purchase amount compared to female customers. Both groups show variation in their purchase amounts, along with several outliers. The distributions also overlap, indicating that although male customers tend to have higher purchase amounts, there is considerable similarity between the two groups.

Recommendations:

- Walmart should maintain sufficient stock and provide attractive offers for Product Category 1, as it has the highest spending across all three city categories.
- Walmart should improve the promotion and visibility of product categories with lower or moderate spending so that they can attract more customers and potentially become higher-performing categories in the future.
- Walmart should review Product Category 19 and consider improving its pricing, promotion, or product selection to increase customer interest and sales.
- Walmart should focus more Black Friday promotions and product availability in City B, as it currently has the highest total spending.
- Walmart should increase promotional activities in City C because it has a large customer base but lower average spending per customer. Increasing spending from existing customers could help City C become a stronger-performing city.

- Walmart should attract more customers in City A because it has the highest average spending per customer. Increasing its customer base while maintaining the current spending level could help increase its overall sales.
- Walmart should provide special offers to customers from Occupation 4, as it has the highest overall spending. Walmart should also encourage customers from lower-spending occupations through suitable discounts and offers.
- Walmart should focus more on customers aged 26–35 because this group has the highest total spending. Walmart can also create suitable offers for older customers, as their average purchase amount shows an increasing trend with age.
- Walmart should create different offers for male and female customers based on the products they purchase, as the effect of product categories on purchase amount differs between the two groups.
- Walmart should not make marital status the main factor when targeting customers, as married and single customers have almost the same average purchase amount.
- Walmart should create targeted offers for customers based on how long they have stayed in their current city, especially for customers who have stayed for 1 year, as this group has the highest total spending.
- Walmart should regularly identify cities, occupations, age groups, and product categories that are performing below the top level but show potential for growth, and invest in promotions and customer engagement to help these segments become stronger contributors to sales.